Major Assignment

Machine Learning

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Contents

Comparative Analysis of Preprocessing Pipelines, Regularization Techniques, and Class Imbalance Handling for Epileptic Seizure Prediction Using Logistic Regression

Abstract

Epilepsy is a neurological disorder characterized by abnormal electrical activity in the brain that can lead to seizures. Machine learning techniques have been widely used to improve seizure prediction and classification. This study investigates the impact of preprocessing pipelines, regularization techniques, and class imbalance handling methods on the performance of Logistic Regression models for epileptic seizure prediction. Three seizure-related datasets were used to evaluate the effectiveness of two preprocessing pipelines and three regularization methods: L1, L2, and Elastic Net. Model performance was evaluated using Accuracy, F1-score, and Precision-Recall Area Under Curve (PR-AUC). Experimental results demonstrate that preprocessing order significantly affects model performance and that L1 regularization provides effective feature selection while maintaining competitive predictive performance.

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1. Introduction

Epilepsy affects millions of people worldwide and is one of the most common neurological disorders. Electroencephalogram (EEG) signals are widely used to analyze brain activity and identify seizure events. Machine learning techniques can assist healthcare professionals by automatically detecting seizure patterns from EEG data.

The performance of machine learning models depends not only on the learning algorithm but also on data preprocessing, regularization, and class imbalance handling strategies. Therefore, this study focuses on investigating how these factors influence the generalization ability of Logistic Regression models in seizure prediction tasks.

The objectives of this study are:

1. Compare different preprocessing pipelines.
2. Evaluate Logistic Regression as a baseline model.
3. Demonstrate underfitting and overfitting scenarios.
4. Compare L1, L2, and Elastic Net regularization techniques.
5. Analyze feature sparsity produced by regularization.
6. Investigate class imbalance handling methods.
7. Determine which techniques provide the best generalization performance.

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2. Datasets

Three seizure-related datasets were used in this study.

Dataset 1: Epileptic Seizure Recognition Dataset

This dataset contains EEG recordings represented by 178 numerical features. The original multiclass problem was converted into a binary classification task where seizure activity was labeled as class 1 and non-seizure activity was labeled as class 0.

Characteristics:

- Number of Features: 178
- Type: EEG Feature Dataset
- Classification Task: Binary Classification
- High-dimensional feature space

Dataset 2: BEED_Data Dataset

This dataset contains EEG signal measurements represented by 16 numerical features and one target variable.

Characteristics:

- Number of Features: 16
- EEG signal-based dataset
- Lower dimensional feature space

Dataset 3: EEG Seizure Analysis Dataset

This dataset contains EEG signal values represented using 16 numerical features.

Characteristics:

- Number of Features: 16
- EEG signal-based dataset
- Similar structure to Dataset 2

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3. Data Preprocessing

Two preprocessing pipelines were developed and compared.

Pipeline A

Pipeline A consisted of:

1. Standardization using StandardScaler

2. Feature Selection using SelectKBest
3. Logistic Regression Classification

Workflow:

Raw Data → Standardization → Feature Selection → Logistic Regression

The purpose of feature selection was to remove irrelevant features and retain only the most informative attributes.

Pipeline B

Pipeline B consisted of:

1. Min-Max Scaling
2. Principal Component Analysis (PCA)
3. Logistic Regression Classification

Workflow:

Raw Data → Scaling → PCA → Logistic Regression

PCA was applied to reduce dimensionality while preserving 95% of the variance.

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4. Logistic Regression Model

Logistic Regression was selected as the baseline classifier due to its simplicity, interpretability, and support for regularization techniques.

The Logistic Regression probability function is:

$P(y=1|x)=1/(1+e^{-(\beta_0+\beta^T x)})$

Advantages of Logistic Regression:

- Computational efficiency
- Interpretability
- Support for L1, L2, and Elastic Net regularization
- Strong baseline for binary classification

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5. Regularization Study

Three regularization methods were evaluated.

5.1 L1 Regularization (Lasso)

L1 regularization adds an absolute-value penalty to the loss function.

Advantages:

- Produces sparse models
- Performs automatic feature selection
- Eliminates irrelevant features

5.2 L2 Regularization (Ridge)

L2 regularization penalizes the squared magnitude of coefficients.

Advantages:

- Reduces model variance
- Retains all features
- Improves stability

5.3 Elastic Net

Elastic Net combines L1 and L2 penalties.

Advantages:

- Balances feature selection and stability
- Suitable for high-dimensional datasets
- Reduces overfitting

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6. Experimental Results

6.1 Pipeline A Results

Dataset	Regularization	Accuracy	F1-Score	PR-AUC
Dataset 1	L1	0.8109	0.7360	0.4748
Dataset 1	L2	0.8104	0.7351	0.4753
Dataset 1	Elastic Net	0.8104	0.7351	0.4754
Dataset 2	L1	0.4575	0.4643	N/A
Dataset 2	L2	0.4544	0.4609	N/A
Dataset 2	Elastic Net	0.4556	0.4624	N/A
Dataset 3	L1	0.4575	0.4643	N/A
Dataset 3	L2	0.4544	0.4609	N/A
Dataset 3	Elastic Net	0.4556	0.4624	N/A

6.2 Pipeline B Results

Dataset	Regularization	Accuracy	F1-Score	PR-AUC
Dataset 1	L1	0.8030	0.7183	0.4587
Dataset 1	L2	0.8043	0.7213	0.4534
Dataset 1	Elastic Net	0.8030	0.7183	0.4563
Dataset 2	L1	0.4269	0.4230	N/A
Dataset 2	L2	0.4394	0.4379	N/A
Dataset 2	Elastic Net	0.4331	0.4311	N/A
Dataset 3	L1	0.4269	0.4230	N/A
Dataset 3	L2	0.4394	0.4379	N/A
Dataset 3	Elastic Net	0.4331	0.4311	N/A

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7. Sparsity Analysis

Feature sparsity was evaluated by counting the number of coefficients reduced to zero.

Regularization	Zero Weights	Total Weights	Sparsity (%)
L1	5	50	10.0
L2	0	50	0.0

Analysis:

L1 regularization eliminated 10% of the model coefficients, demonstrating its ability to perform automatic feature selection. In contrast, L2 regularization retained all coefficients while reducing their magnitudes. This confirms the feature-selection capability of L1 regularization.

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8. Underfitting and Overfitting Analysis

Underfitting Scenario

Training Accuracy = 0.5002

Testing Accuracy = 0.4944

The training and testing accuracies are both low and nearly identical, indicating that the model failed to learn important patterns from the data. This behavior is characteristic of underfitting and high bias.

Overfitting Scenario

Training Accuracy = 0.4817

Testing Accuracy = 0.4544

The model exhibited poor generalization performance and unstable behavior. The decrease in testing accuracy indicates that increasing model complexity did not improve predictive performance.

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9. Class Imbalance Handling

SMOTE was applied to address class imbalance.

SMOTE Results

Method	Accuracy	F1-Score	PR-AUC
SMOTE	0.4331	0.4311	0.3862

Analysis:

The application of SMOTE did not improve classification performance. The synthetic samples generated by SMOTE may have introduced additional noise, reducing the model's ability to distinguish between classes effectively.

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10. Comparative Analysis

Does preprocessing order affect results?

Yes. Pipeline A consistently outperformed Pipeline B across all datasets.

For Dataset 1:

- Pipeline A Accuracy: 81.09%
- Pipeline B Accuracy: 80.43%

This indicates that Standardization followed by Feature Selection preserved more useful information than PCA-based dimensionality reduction.

Which regularization generalizes best across datasets?

L1 regularization achieved the highest Accuracy and F1-score on Dataset 1 while also providing feature selection capabilities. Therefore, L1 demonstrated the best overall balance between performance and interpretability.

Does Elastic Net consistently outperform L1 and L2?

No. Elastic Net achieved slightly higher PR-AUC values but did not consistently outperform L1 or L2 in terms of Accuracy and F1-score. The differences among the three methods were minimal.

How does imbalance handling interact with regularization?

The SMOTE experiment did not improve model performance. This suggests that regularization alone was more beneficial than synthetic oversampling for the evaluated datasets.

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11. Conclusion

This study investigated the impact of preprocessing pipelines, regularization techniques, and class imbalance handling methods on epileptic seizure prediction using Logistic Regression.

The results showed that preprocessing order significantly affects model performance. Pipeline A, consisting of Standardization followed by Feature Selection, consistently outperformed Pipeline B, which used PCA-based dimensionality reduction.

Among the evaluated regularization methods, L1 regularization achieved the highest overall Accuracy and F1-score while also producing sparse models through automatic feature selection. Elastic Net provided comparable performance but did not consistently outperform L1 or L2.

Sparsity analysis confirmed that L1 regularization eliminated unnecessary features, improving model interpretability. Additionally, class imbalance handling using SMOTE did not significantly improve classification performance.

Overall, the experiments demonstrate the importance of preprocessing design and regularization strategy selection in seizure prediction systems.
